

CSE370 Spring 2026

Assignment: **03**

Total Marks: **20**

Submission Instructions:

1. Write your Full Name, Student-ID, Theory Section on top of the first page
 2. Your answer should be **handwritten**
 3. Submit online here: <https://forms.gle/FFjzk7ABMrT3kZwY8>
 4. **NO LATE SUBMISSION WILL BE ACCEPTED.**
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Question 1 (10 Points)

Construct a B+ tree of order $n = 4$ for the following search key values inserted in the given order:

**Lily, Howard, Bernadette, Amy, Monica,
Marshall, Robin, Sheldon, Gunther, Penny,
Barney, Phoebe, Ted, Joey, Chandler.**

Each time there is a split, **a new B+ tree must** be drawn.

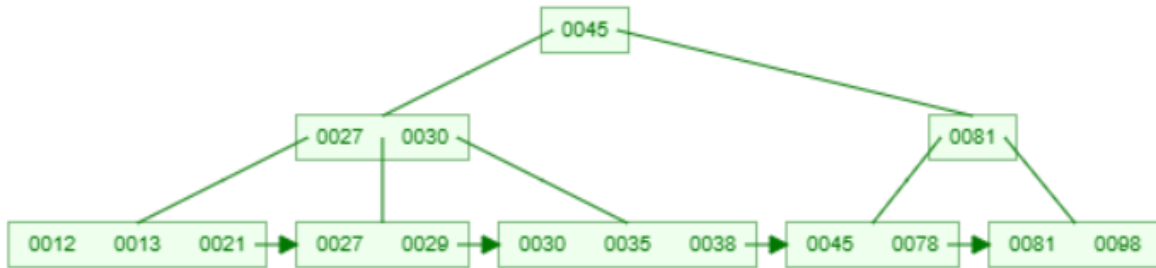
Use numerical mapping:

**Lily 8, Howard 6, Bernadette 3, Amy 1, Monica 10,
Marshall 9, Robin 13, Sheldon 14, Gunther 5, Penny 11,
Barney 2, Phoebe 12, Ted 15, Joey 7, Chandler 4.**

Insert 8, 6, 3, 1, 10, 9, 13, 14, 5, 11, 2, 12, 15, 7, 4.

Question 2 (8 Points)

Consider the following, B+ Tree of order $n = 4$.



- What is the minimum number of keys allowed for each node?
- Draw the B+ Tree after each of the following commands.
 - Del 38
 - Del 45
 - Del 30
 - Del 35
 - Del 78
 - Del 98
 - Del 81

Question 3 (2 Points)

Consider the following table below. Here **phone_id** and **brand** are indexes.

Identify and **explain** which one is the primary index and which is the secondary index.

phone_id	brand	price(k)	color
1102	Apple	105	Green
1106	Apple	115	Black
1101	Google	60	Black
1104	Google	80	Red
1109	Honor	40	Blue
1103	Honor	30	Red
1107	Motorola	45	Green
1105	Nokia	10	Grey
1108	Oneplus	95	White
1110	Xiaomi	20	Navy